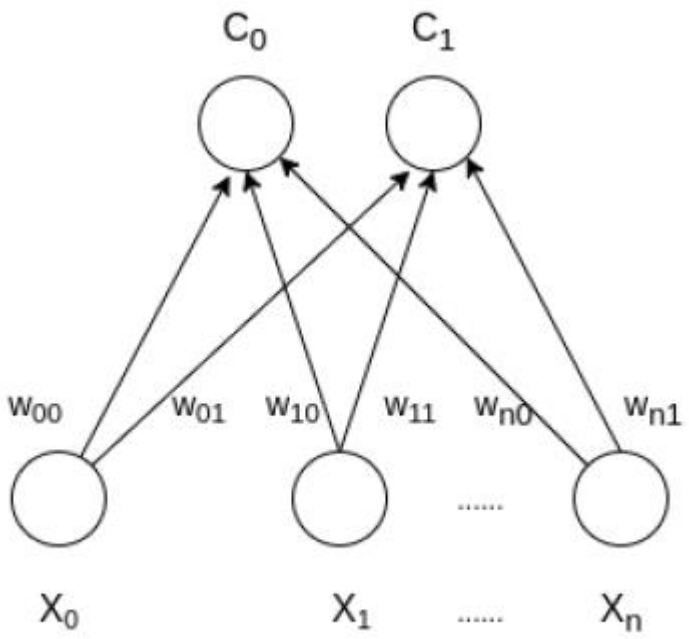
Self Organizing Maps – Kohonen Maps

Self Organizing Map (or Kohonen Map or SOM) is a type of Artificial Neural Network which is also inspired by biological models of neural systems from the 1970s. It follows an unsupervised learning approach and trained its network through a competitive learning algorithm. SOM is used for clustering and mapping (or dimensionality reduction) techniques to map multidimensional data onto lower-dimensional which allows people to reduce complex problems for easy interpretation. SOM has two layers, one is the Input layer and the other one is the Output layer. The architecture of the Self Organizing Map with two clusters and n input features of any sample is given below:



How do SOM works?

Let's say an input data of size (m, n) where m is the number of training examples and n is the number of features in each example. First, it initializes the weights of size (n, C) where C is the number of clusters. Then iterating over the input data, for each training example, it updates the winning vector (weight vector with the shortest distance (e.g Euclidean distance) from training example). Weight updation rule is given by :

$w_{ij} = w_{ij}(old) + alpha(t) * (x_i^k - w_{ij}(old))$

where alpha is a learning rate at time t, j denotes the winning vector, i denotes the ith feature of training example and k denotes the kth training example from the input data. After training the SOM network, trained weights are used for clustering new examples. A new example falls in the cluster of winning vectors.

Algorithm

Training:

Step 1: Initialize the weights w_{ij} random value may be assumed. Initialize the learning rate α.

Step 2: Calculate squared Euclidean distance.

$$D(j) = \Sigma\ (wij - xi)^2 \quad \text{where } i=1 \text{ to } n \text{ and } j=1 \text{ to } m$$

Step 3: Find index J, when D(j) is minimum that will be considered as winning index.

Step 4: For each j within a specific neighborhood of j and for all i, calculate the new weight.

$$wij(new)=wij(old) + \alpha[xi - wij(old)]$$

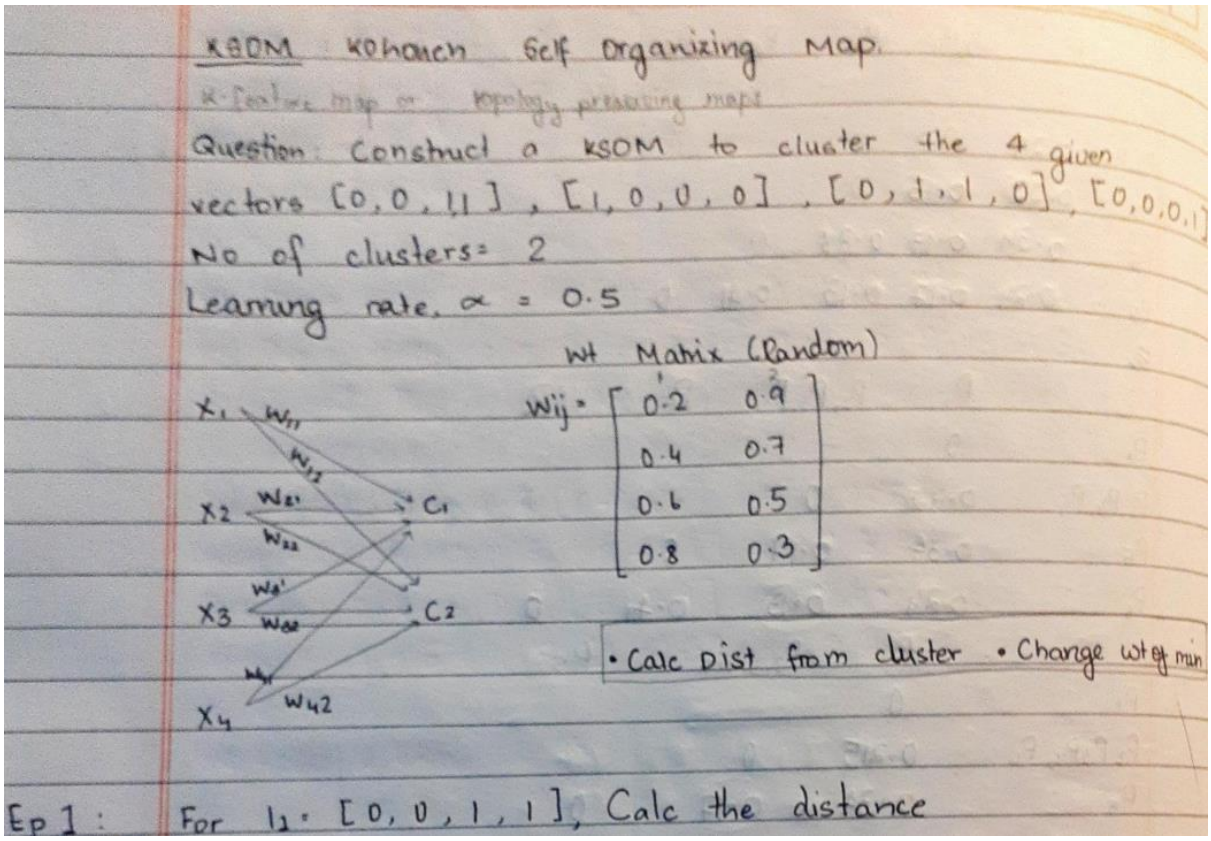
Step 5: Update the learning rule by using :

$$\alpha(t+1) = 0.5 * t$$

Step 6: Test the Stopping Condition.

Numerical example:-

Stopping criteria is mentioned in the question itself. (No of epoch should be mentioned in the starting otherwise for every consecutive epoch learning rate will always reduce to half.



$$D_{11} = \Sigma(w_{ij} - x_1)^2$$
$$= (0.2-0)^2 + (0.4-0)^2 + (0.6-1)^2 + (0.8-1)^2$$
$$= 0.4$$
$$D_{12} = (0.9-0)^2 + (0.7-0)^2 + (0.5-1)^2 + (0.3-1)^2$$
$$= 2.04$$

$D_{11} < D_{12}$ so updates wts for c_1

$$W_{ij(new)} \; W_{ij(old)} + \alpha [X_1 - W_{ij(old)}]$$

$$W_{11(new)} = 0.2 + 0.5 [0 - 0.2]$$
$$= 0.1$$
$$W_{21(new)} = 0.4 + 0.5 [0 - 0.4]$$
$$= 0.2$$
$$W_{31(new)} = 0.6 + 0.5 [1 - 0.6]$$
$$= 0.8$$

$$W_{41} = 0.8 + 0.5 [1 - 0.8]$$
$$= 0.9$$

$$W = \begin{bmatrix} 0.1 & 0.9 \\ 0.2 & 0.7 \\ 0.8 & 0.5 \\ 0.9 & 0.3 \end{bmatrix}$$

$$D_{21} = (0.1-1)^2 + (0.2-0)^2 + (0.8)^2 + 0.9^2 = 2.3$$
$$D_{22} = (0.9-1)^2 + 0.7^2 + 0.5^2 + 0.3^2 = 0.84$$

$D_{22} < D_{21}$ so update C_2

$$W_{12} = 0.9 + 0.5 [1 - 0.9] = 0.95$$
$$W_{22} = 0.7 + 0.5 [0 - 0.7] = 0.35$$

$W_{32} = 0.5 + 0.5\,[0 - 0.5] = 0.25$

$W_{42} = 0.3 + 0.5\,[0 - 0.3] = 0.15$

$$W = \begin{bmatrix} 0.1 & 0.95 \\ 0.2 & 0.35 \\ 0.8 & 0.25 \\ 0.9 & 0.15 \end{bmatrix}$$

$D_{31} = 0.1^2 + (0.2-1)^2 + (0.8-1)^2 + 0.9^2 = 1.5$

$D_{32} = 0.95^2 + (0.35-1)^2 + (0.25-1)^2 + 0.15^2 = 1.91$

$D_{31} < D_{32}$ so update C_1

$W_{11} = 0.1 + 0.5[0 - 0.1] = 0.05$

$W_{21} = 0.2 + 0.5[1 - 0.2] = 0.6$

$W_{31} = 0.8 + 0.5[1 - 0.8] = 0.9$

$W_{41} = 0.9 + 0.5[0 - 0.9] = 0.45$

$$W = \begin{bmatrix} 0.05 & 0.95 \\ 0.6 & 0.35 \\ 0.9 & 0.25 \\ 0.45 & 0.15 \end{bmatrix}$$

$D_{41} = 0.05^2 + 0.6^2 + 0.9^2 + (0.45-1)^2 = 1.475$

$D_{42} = 0.95^2 + 0.35^2 + 0.25^2 + (0.15-1)^2 = 1.81$

$D_{41} < D_{42}$ update C_1

$W_{11} = 0.05 + 0.5\,[0 - 0.05] = 0.025$

$W_{21} = 0.6 + 0.5\,[0 - 0.6] = 0.3$

$W_{31} = 0.9 + 0.5\,[0 - 0.9] = 0.45$

$W_{41} = 0.45 + 0.5\,[1 - 0.45] = 0.725$

$$W = \begin{bmatrix} 0.025 & 0.95 \\ 0.3 & 0.35 \\ 0.45 & 0.25 \\ 0.725 & 0.15 \end{bmatrix}$$

Ep 2: $\alpha(t+1) = 0.5\alpha(t)$ unless mentioned otherwise

$= 0.5 \times 0.5$

$= 0.25$

Answer after Ep: Updated Wt Matrix

Cluster

$C_1 = \{V_1, V_3, V_4\}$ $C_2 = \{V_2\}$